Foreign Direct Investment and Business Investment in Canada

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October 2026

Abstract

Foreign direct investment (FDI) inflows are not significantly associated with business investment in the main US-dollar models for Canada over 1981–2018. Using annual data from Statistics Canada and the World Bank, I estimate a levels equation with autoregressive errors and an error-correction model that relates investment changes to expenditure, capacity slack, and FDI. The FDI coefficient is positive but statistically insignificant in the main US-dollar specifications. Canadian-dollar equations produce a negative association of approximately −0.107 to −0.108, including after controlling for the exchange-rate level. The contrast shows that the result depends on currency denomination and model specification. A cointegrating relationship between investment and expenditure is supported in the main US-dollar sample, while the main ECM still exhibits residual serial correlation. The estimates provide no robust evidence that FDI crowds in business investment. They also do not establish that foreign inflows cause domestic investment to decline. The distinction matters because FDI measures financial inflows, which can include acquisitions of existing assets, rather than additions to productive capacity alone.

1 Introduction

Foreign direct investment can finance new plants, equipment, and technology, but a recorded inflow need not create an equivalent amount of fixed capital. Foreign investors may acquire an existing firm or provide financing that substitutes for funds available elsewhere. Domestic firms may respond by expanding production, supplying foreign affiliates, or reducing investment in markets where competition increases. Whether FDI adds to business investment is therefore an empirical question about the response of capital formation, not simply the volume of foreign funds received.

I examine this question for Canada using annual observations from 1981 to 2018. The dependent variable is business gross fixed capital formation, and the explanatory variables are gross national expenditure, estimated capacity slack, and FDI net inflows. Expenditure captures demand conditions, while the capacity measure allows investment to respond to the economy's position relative to estimated potential output. The investment series covers the business sector as a whole, including foreign-owned firms. It therefore measures investment undertaken in Canada rather than investment by domestic owners alone.

The analysis revisits an earlier levels regression that reported a positive FDI coefficient and interpreted its size relative to one as evidence about complementarity. I reconstruct the dataset and examine the assumptions needed for that interpretation. The distinction between a trending relationship in levels and a short-run investment response is particularly important in a small annual sample. I therefore test the series for unit roots and examine cointegration before estimating an error-correction model.

The main US-dollar models produce positive but insignificant FDI estimates. Canadian-dollar equations instead show a negative association, which remains after including the exchange-rate level. I examine this difference through alternative currency specifications, a sample extension through 2024, and an interaction between FDI and capacity slack. These comparisons indicate how much the finding depends on the specification; they do not identify an exogenous change in foreign investment.

2 Background and investment model

Agosin and Mayer (2000) examine whether FDI crowds domestic investment in or out in developing economies over 1970–1996. Their regional estimates show crowding in in Asia and crowding out in Latin America. Borensztein, De Gregorio, and Lee (1995), using data for 69 developing economies, report a more-than-proportional relationship between FDI and total investment. These findings suggest that the investment response depends on the receiving economy and the empirical specification.

Van Loo (1977) examines the same question for Canada and provides the reference for the earlier version of this study. Van Loo's reported FDI coefficient of about 1.40 exceeds one, whereas the main US-dollar estimates here are below 0.14; differences in sample, measurement, and model specification prevent attributing this gap to a historical change in the investment response.

$$I_t = \alpha + \beta Y_t + \gamma U_t + \delta FDI_t + \varepsilon_t \quad (1)$$

I denotes business gross fixed capital formation, Y gross national expenditure, and U estimated capacity slack. Higher expenditure is expected to increase investment, while greater unused capacity is expected to reduce it. The expected signs are therefore $\beta > 0$ and $\gamma < 0$. The sign of δ is an empirical question. I also test whether FDI has a stronger investment association when capacity is available: near full capacity, additional demand may raise prices rather than expand production. Since U measures slack, this hypothesis predicts a positive FDI × U coefficient.

Two measurement issues affect interpretation. Expenditure includes investment, creating an accounting relationship with the dependent variable. FDI is a financial-flow measure and need not correspond to new fixed capital formation in the same year. As Agosin and Mayer (2000) note, acquisitions transfer ownership, while foreign firms can also finance investment domestically. A coefficient below one does not by itself establish displacement of domestic firms.

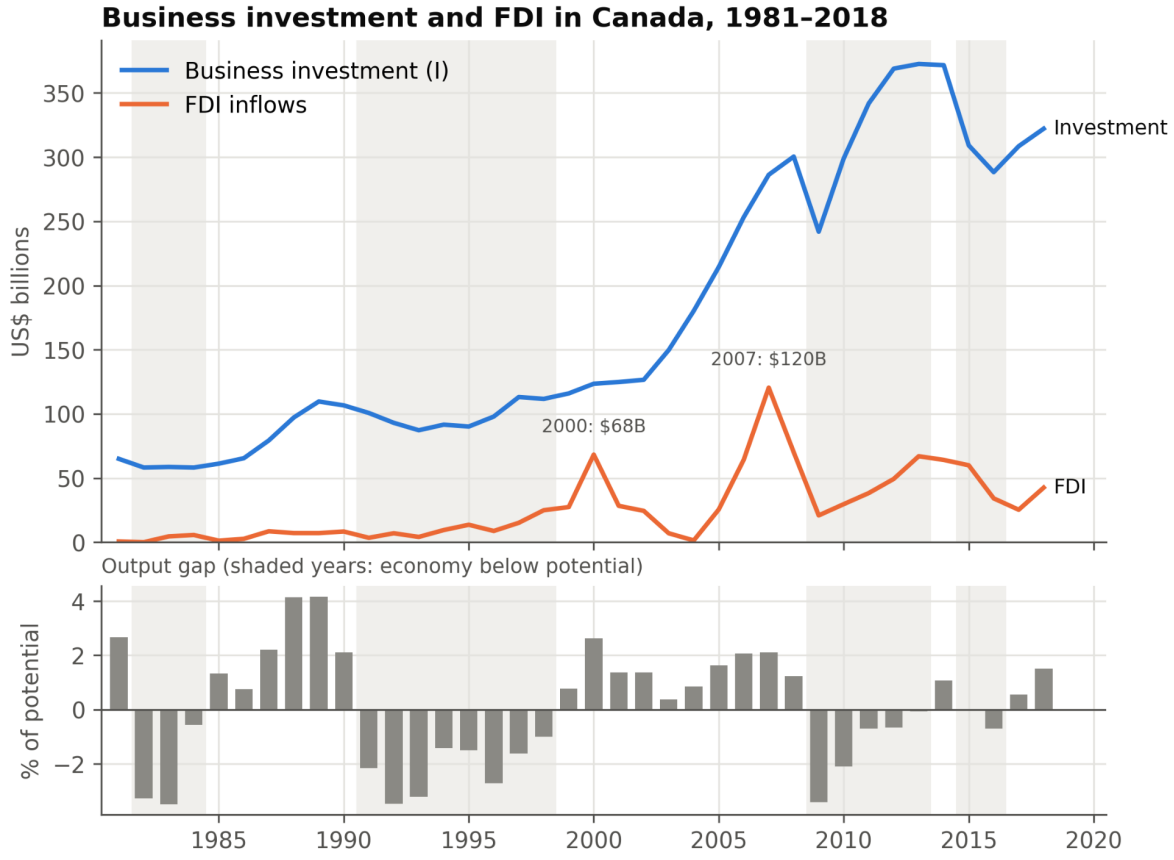
3 Data and methods

3.1 Data

The main sample contains 38 annual observations for 1981–2018, with an extension through 2024. Statistics Canada Table 36-10-0222-01 supplies business gross fixed capital formation, nominal GDP, and real GDP. World Development Indicators supplies gross national expenditure (NE.DAB.TOTL.CD), FDI net inflows (BX.KLT.DINV.CD.WD), and the annual average exchange rate (PA.NUS.FCRF).

Monetary variables are measured in current-price billions. Canadian-dollar investment and capacity slack are divided by CAD per USD to obtain US-dollar values. World Bank expenditure and FDI are converted to Canadian dollars using the same rate. No estimation variables are missing in the main sample. The earlier paper reports 36 observations; the reconstructed sample contains 38. Differencing leaves 37 observations, and an additional lag leaves 36.

Figure 1 Business investment and FDI inflows



Sources: Statistics Canada Table 36-10-0222-01; World Bank WDI. Output gap from HP filter (λ = 100) on log real GDP.

3.2 Capacity slack

I estimate potential output by applying an HP filter to log real GDP over the available full series, with a smoothing parameter of 100. The log cycle is converted to a proportional gap and then valued at the current GDP price scale. In equation (2), τ denotes the HP trend in log real GDP.

$$c_t = ln(GDP_{t,real}) - \tau_t \tag{2}$$

$$g_t = exp(c_t) - 1 \tag{3}$$

$$U_t = GDP_{t,nominal} \times [\frac{1}{1 + g_t} - 1] \tag{4}$$

Positive U indicates output below estimated potential. This measure differs from the earlier paper's stated definition of potential output minus gross national expenditure. Although the HP cycle removes an estimated trend, U is expressed on a changing nominal GDP scale, so its

stationarity must be tested. Using the full series also makes the estimate sensitive to later observations and data revisions.

3.3 Estimation

The models are estimated in Python using statsmodels. I first estimate the levels equation by OLS and then fit AR(1) errors using iterative GLSAR. This differs from the maximum-likelihood estimator in the earlier EViews output. An autoregressive error specification addresses persistence in the residuals but does not resolve nonstationarity in the underlying series.

Augmented Dickey–Fuller tests include a constant and trend in levels and a constant in first differences. Lag length is selected by AIC. The US-dollar results support treating I and Y as integrated of order one and U and FDI as stationary under the tested specifications. I then test for cointegration between I and Y using the Engle–Granger procedure. The null of no cointegration is rejected in the main US-dollar sample (p = 0.002).

$$I_t = a + bY_t + v_t \tag{5}$$

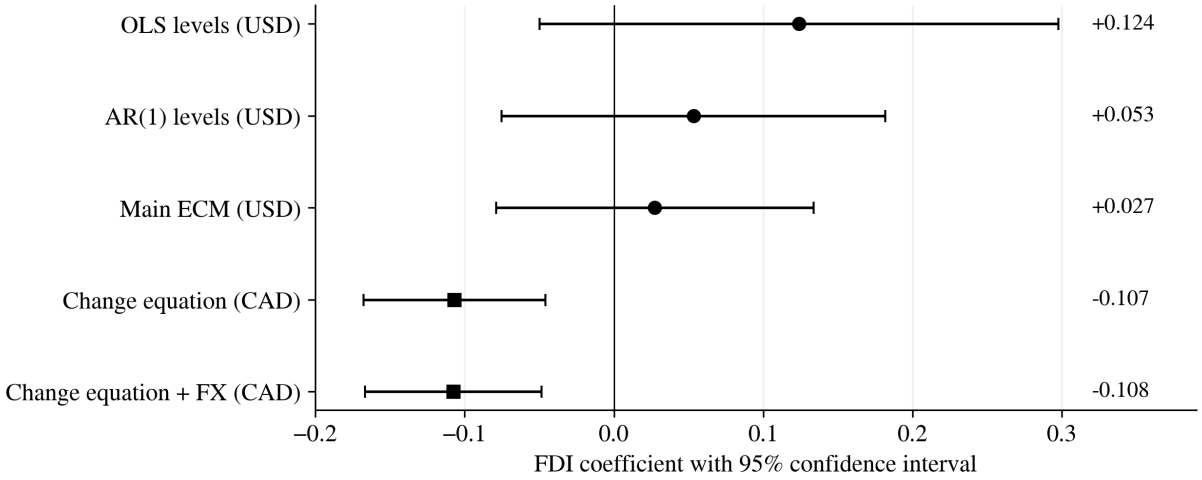
$$ECT_{t-1} = I_{t-1} - \hat{a} - \hat{b}Y_{t-1} \tag{6}$$

$$\Delta I_t = \kappa + \theta \Delta Y_t + \gamma U_t + \delta FDI_t + \lambda ECT_{t-1} + e_t \tag{7}$$

In equation (6), ECT is the previous year's deviation from the estimated I–Y relationship. FDI and U enter the short-run equation in levels; they are not part of the estimated cointegrating vector. A negative λ indicates adjustment toward that relationship. Equation (7) is estimated using HAC standard errors with three lags. Additional specifications examine lagged changes, an FDI–U interaction, and a 2020 indicator in the extended sample.

4 Results

Figure 2 FDI coefficients across specifications



Notes: Bars show 95% confidence intervals. Levels models use conventional standard errors; the main ECM and CAD change equations use HAC errors with three lags. FDI enters in levels throughout. Source: Author's calculations.

The ADF tests reject a unit root for U and FDI but not for I or Y at 5 percent (Table 1). The first-difference tests reject the null for all four variables. These findings motivate the ECM specification, although the short sample limits the power of the tests.

Table 1. Augmented Dickey-Fuller tests, 1981-2018

Variable	ADF stat (level)	p (level)	ADF stat (diff.)	p (diff.)
I	-2.99	0.136	-4.59	<0.001
Y	-3.33	0.061	-4.15	<0.001
U	-3.60	0.029	-5.41	<0.001
FDI	-4.99	<0.001	-5.25	<0.001

Levels include a constant and trend; differences include a constant. Lag length by AIC.

Table 2. Main US-dollar estimates

	OLS levels	AR(1) levels	ECM (HAC)
Y	0.194***	0.198***	
	(0.004)	(0.005)	
ΔY			0.224***
			(0.010)
U	-0.281***	-0.342***	-0.174*
	(0.096)	(0.078)	(0.099)
FDI	0.124	0.053	0.027
	(0.085)	(0.063)	(0.054)
ECT			-0.369**
			(0.158)
Intercept	-11.980***	-16.757***	-2.592*
	(2.952)	(5.784)	(1.345)
R-squared	0.994	0.983	0.923
R-squared Adj.	0.993	0.981	0.914
N	38	37	37

Standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. ECM uses HAC (3 lags).

The estimated FDI coefficient falls from 0.124 in OLS to 0.053 after the AR(1) correction. Neither estimate is significant at 5 percent. In the main ECM, the coefficient is 0.027 (HAC p = 0.615). The US-dollar models therefore provide no significant evidence of an FDI association conditional on the included regressors.

The main ECM adjustment coefficient is −0.369 (p = 0.019). Holding the short-run regressors fixed, approximately 37 percent of the preceding year's deviation from the estimated I–Y relationship is corrected within a year. Expenditure changes are positively associated with investment changes, with an estimated coefficient of 0.224. The capacity-slack coefficient is negative, −0.174, with HAC p = 0.078.

5 Robustness and limitations

Table 3. Sensitivity of the FDI estimate

Specification	N	Coint. p	FDI coef.	FDI p	FDI x U
ECM, US$ (main)	37	0.002	0.027	0.615	-
ECM + FDI x U	37	0.002	0.071	0.356	0.003 (p=0.267)
ECM, C$	37	0.382	-0.071	0.009	-
ECM, US$ to 2024 + COVID	43	0.358	0.032	0.548	-
First differences, C$	37	0.382	-0.107	<0.001	-
First diff., C$ + exch. rate	37	0.382	-0.108	<0.001	-

All models use HAC standard errors (3 lags). Coint. p = Engle-Granger test of I and Y.

Removing ECT from the CAD equation yields an FDI coefficient of −0.107 (p < 0.001). Adding the exchange-rate level leaves it at −0.108 (p < 0.001). Since FDI enters in levels, this associates $1 billion higher annual inflows with a $0.108 billion smaller annual change in business investment. It is not the response to a $1 billion year-to-year increase in FDI.

The exchange-rate control does not explain the CAD–USD difference. A time-varying currency conversion changes the series' dynamics, which an additive exchange-rate term does not reproduce. The CAD variables retained in levels also require separate stationarity tests. Currency conversion should therefore be treated as a specification change rather than a simple rescaling.

The two-lag Breusch–Godfrey test on the main 37-observation ECM rejects the null of no serial correlation (p = 0.001). HAC errors do not correct omitted dynamics or endogeneity. Business investment also includes foreign-owned firms, expenditure contains investment, and the HP-based gap depends on the data vintage. These features prevent the estimates from identifying displacement of domestic investment. Further analysis should test the CAD series and examine parsimonious lag structures and alternative capacity measures.

6 Conclusion

The main US-dollar models provide no significant evidence of an FDI association with business investment in Canada over 1981–2018. The Canadian-dollar change equations produce negative coefficients of approximately −0.107 and −0.108, with the latter including an exchange-rate control. The negative relationship is therefore present in those equations but is not consistent across currency specifications. The main ECM also retains serial correlation, which limits the conclusions that can be drawn from its short-run dynamics.

These results caution against treating aggregate foreign inflows as a direct measure of new capital formation. The economic contribution of FDI depends on the projects financed and the response of other firms. For policy, the relevant distinction is between inflows that expand productive capacity and transactions that mainly change ownership or financing. The aggregate estimates do not separate those channels. Investment data by ownership and FDI entry mode would help determine whether the negative Canadian-dollar association reflects displacement, acquisition activity, or common economic conditions.

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